

MATH 411: Week 6

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Problem 8.9

Suppose that f and g are functions defined on $[a, b]$ and are both differentiable at a point $x \in [a, b]$. Show that if $g(x) \neq 0$,

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

Let f and g be defined on $[a, b]$ and differentiable at some $x \in [a, b]$, with $g(x) \neq 0$. We'll show that

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

We first observe that $(g^{-1})'(x)$ is given by

$$\begin{aligned}(g^{-1})'(x) &= \lim_{t \rightarrow x} \frac{\frac{1}{g(t)} - \frac{1}{g(x)}}{t - x} \\&= \lim_{t \rightarrow x} \frac{\frac{g(x) - g(t)}{g(t)g(x)}}{t - x} \\&= \lim_{t \rightarrow x} \left(\frac{1}{g(t)g(x)} \cdot \frac{g(x) - g(t)}{t - x} \right) \\&= \lim_{t \rightarrow x} \frac{1}{g(t)g(x)} \cdot \lim_{t \rightarrow x} \frac{g(x) - g(t)}{t - x} \\&= \frac{1}{g(x)^2} (-g'(x)) \\&= -\frac{g'(x)}{g(x)^2},\end{aligned}$$

since Problem 8.6 implies g is continuous at x and hence that $t \mapsto \frac{1}{g(t)g(x)}$ is

continuous at x , since $g(x) \neq 0$. Problem 8.8 now ensures

$$\begin{aligned} \left(\frac{f}{g}\right)'(x) &= (fg^{-1})'(x) = f(x)(g^{-1})'(x) + f'(x)g^{-1}(x) \\ &= -f(x)\frac{g'(x)}{g(x)^2} + f'(x)\frac{1}{g(x)} \\ &= -f(x)\frac{g'(x)}{g(x)^2} + f'(x)\frac{g(x)}{g(x)^2} \\ &= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}. \end{aligned}$$

□

Problem 8.13

Let f and g be continuous real-valued functions on $[a, b]$ which are differentiable on (a, b) . Define, for all $t \in [a, b]$,

$$h(t) = [f(b) - f(a)]g(t) - [g(b) - g(a)]f(t).$$

We first show that h is continuous on $[a, b]$ and differentiable on (a, b) . Problem 6.6 ensures h is continuous on $[a, b]$. $t \mapsto f(b) - f(a)$, $t \mapsto g(b) - g(a)$, g , and f are all differentiable on (a, b) , so Problems 8.8 and 8.7 ensure h is also differentiable on (a, b) .

Now we show that $h(a) = h(b)$:

$$\begin{aligned} h(a) &= [f(b) - f(a)]g(a) - [g(b) - g(a)]f(a) \\ &= f(b)g(a) - f(a)g(a) - [g(b)f(a) - g(a)f(a)] \\ &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - g(b)f(a) \\ &= g(b)f(b) - g(b)f(a) - g(b)f(b) + f(b)g(a) \\ &= [f(b) - f(a)]g(b) - [g(b) - g(a)]f(b) \\ &= h(b). \end{aligned}$$

Finally, suppose $h'(x) = 0$ for some $x \in (a, b)$. We'll show that

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x).$$

Since $t \mapsto f(b) - f(a)$ and $t \mapsto g(b) - g(a)$ are constant functions, Problems 8.7 and 8.8 ensure

$$h'(t) = [f(b) - f(a)]g'(t) - [g(b) - g(a)]f'(t).$$

So we have:

$$\begin{aligned} 0 &= h'(x) = [f(b) - f(a)]g'(x) - [g(b) - g(a)]f'(x) \\ [f(b) - f(a)]g'(x) &= [g(b) - g(a)]f'(x). \end{aligned}$$

□

Problem 8.14

Using the conclusions from the previous problem, show that there exists some $x \in (a, b)$ such that $h'(x) = 0$.

Problem 6.13 ensures that h attains a maximum value M and a minimum value m at some $x_{\max}$ and $x_{\min}$, respectively, in $[a, b]$.

If either $x_{\min} \in (a, b)$ or $x_{\max} \in (a, b)$, then Theorem 8.1 ensures either $f'(x_{\min}) = 0$ or $f'(x_{\max}) = 0$. On the other hand, if $x_{\min} = a$ and $x_{\max} = b$, then we have

$$M = f(x_{\max}) = f(a) = f(b) = f(x_{\min}) = m,$$

and if $x_{\max} = b$ and $x_{\min} = a$, then we have

$$m = f(x_{\min}) = f(a) = f(b) = f(x_{\max}) = M.$$

But if $m = M$, then for all $x \in [a, b]$ we have $h(x) \geq m = M$ and $h(x) \leq M$; so $h(x) = M$ and $h'(x) = 0$ for any $x \in [a, b]$. In any case, we've shown that an x with the desired property exists.

□

Problem 8.15

Let f be continuous on $[a, b]$ and differentiable on (a, b) . It follows directly from Cauchy's Mean Value Theorem that there is an $x \in (a, b)$ such that

$$\begin{aligned} [f(b) - f(a)]g'(x) &= (g(b) - g(a))f'(x) \\ [f(b) - f(a)] \cdot 1 &= (b - a)f'(x) \\ f(b) - f(a) &= (b - a)f'(x), \end{aligned}$$

where g is the identity function on $[a, b]$.

Let f be continuous on $[a, b]$, with $f'(x) = 0$ for all $x \in (a, b)$. Let $x, y \in [a, b]$ with $x < y$. Then f is differentiable on (x, y) and continuous on $[x, y]$. By our discussion above, there's a $p \in (x, y)$ such that

$$\begin{aligned} f(y) - f(x) &= (y - x)f'(p) = 0 \\ f(y) &= f(x). \end{aligned}$$

So f has some value over the entirety of (a, b) . f must also have this value at a and b , or else f would have discontinuities there. So we've shown that f is constant.

Let f be continuous on $[a, b]$, with $f'(x) \geq 0$ for all $x \in (a, b)$. Let $x, y \in (a, b)$ with $x < y$. By our discussion above, there's a $p \in (x, y)$ such that

$$\begin{aligned} f(y) - f(x) &= (y - x)f'(p) \geq 0 \\ f(y) &\geq f(x). \end{aligned}$$

Suppose $f(a) > f(p)$ for some $p \in (a, b)$. Then the intermediate value theorem ensures there's a $c \in (a, p) \subseteq (a, b)$ such that $f(p) < f(c) < f(a)$. But we've already shown $f(c) \geq f(p)$, so we conclude $f(a) \leq f(p)$ for all $p \in (a, b)$. A similar argument shows that $f(p) \leq f(b)$ for all $p \in (a, b)$. Transitivity ensures $f(a) \leq f(b)$, since $(a, b) \neq \emptyset$.

□